

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

$\phi(x, y, z) \rightarrow$ scalar field ($\vec{\phi}(x, y, z) \rightarrow$ vector field)

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$= \left(\hat{x} \frac{\partial \phi}{\partial x} + \hat{y} \frac{\partial \phi}{\partial y} + \hat{z} \frac{\partial \phi}{\partial z} \right) \cdot (\hat{x} dx + \hat{y} dy + \hat{z} dz)$$

$$\begin{cases} \vec{r} = \hat{x} x + \hat{y} y + \hat{z} z \\ d\vec{r} = \hat{x} dx + \hat{y} dy + \hat{z} dz \end{cases}$$

$$= \underbrace{\left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right)}_{\vec{\nabla} \phi} \cdot d\vec{r}$$

From here we can calculate that how much our function changes by changing x, y, z .

$\vec{\nabla} \rightarrow$ this is a vector operator which operates on a scalar (ϕ) $\rightarrow$ the result is vector.

direction must be the direction in which the scalar is increasing most rapidly.

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$$\frac{d\phi}{dr} = (\vec{\nabla} \phi) \cdot \hat{n}$$

we can choose $\hat{n}$ and based on that our $\frac{d\phi}{dr}$ will

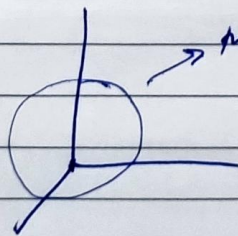
change

directional derivative.

$$\frac{d\phi}{dr} = |\vec{\nabla} \phi| \cdot 1 \cdot \cos \theta \Rightarrow \frac{d\phi}{dr} \text{ will be max. when } \cos \theta = 1$$

$$\left| \frac{d\phi}{dr} \right|_{\max} = |\vec{\nabla} \phi|$$

- The gradient for a level surface is always normal to the surface.
- If we have a heated sphere, then the direction of ~~gradient~~ ^{mass} will be outwards to inwards.



this is heated sphere.

direction of gradient is inwards.

Electrostatic field is a conservative field

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$$\vec{E} = - \frac{d\phi}{dx} \hat{x}$$

$$\text{In 3D: } \vec{E} = - \left(\frac{\partial \phi}{\partial x} \hat{x} + \frac{\partial \phi}{\partial y} \hat{y} + \frac{\partial \phi}{\partial z} \hat{z} \right)$$

$$= - \nabla \phi$$

- * Conservative field is the one which can be written as a gradient of a quantity and for this the path chosen is independent.

* $\vec{V} = \nabla U$ → conservative field.

$$\int_A^B \vec{V} \cdot d\vec{r} = \int_A^B (\nabla U) \cdot d\vec{r} = \int_A^B dU = U(B) - U(A)$$

Flux

Open surface $\Rightarrow$

closed surface $\Rightarrow$

$\vec{\nabla} \cdot \vec{v} \Rightarrow$ divergence

$\vec{\nabla} \times \vec{v} \Rightarrow$ curl of $\vec{v}$

$\vec{v} \Rightarrow$ vector

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$\nabla \rightarrow$ does not have any direction itself (it is an operator)
By $\nabla \phi \rightarrow$ this has direction.

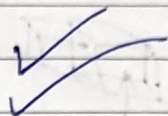
Don't say if $\vec{\nabla} \times \vec{A} = 0$

then these are parallel

and

if $\vec{\nabla} \cdot \vec{A} = 0$

then these are $\perp$



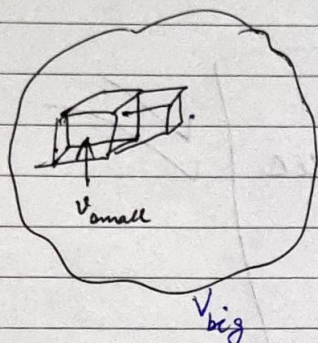
$$\vec{\nabla} = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right)$$

$$\vec{\nabla} \cdot \vec{v} = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot (\hat{x} v_x + \hat{y} v_y + \hat{z} v_z)$$

$$= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \Rightarrow \text{divergence}$$

P.T.D

lect - 3

Divergence theorem

$$\int_{V_{\text{big}}} (\nabla \cdot \vec{v}) dv = \sum_{\substack{\text{all tiny} \\ \text{parallelepiped} \\ dv \rightarrow 0}} \lim (\vec{v} \cdot \vec{v}) dv$$

$$= \sum_{\text{small volume}} \oint \vec{v} \cdot d\vec{s}$$

$$= \sum_{\substack{\text{unshared} \\ \text{faces coinciding} \\ \text{with the boundary} \\ \text{of the big volume}}} \vec{v} \cdot d\vec{s} = \oint_{\text{entire surface}} \vec{v} \cdot d\vec{s}$$

$$\boxed{\int_V (\nabla \cdot \vec{v}) dv = \oint_S \vec{v} \cdot d\vec{s}} \quad \left(\text{Gauss's divergence theorem.} \right)$$

Combining divergence and gradient: Laplacian

$$\nabla \cdot (\nabla U) = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \left(\hat{x} \frac{\partial U}{\partial x} + \hat{y} \frac{\partial U}{\partial y} + \hat{z} \frac{\partial U}{\partial z} \right)$$

$$= \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2}$$

$$\boxed{\nabla^2 U = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) U}$$

↳ can be applied to scalar/vector

gravitational potential ^{also} satisfies Laplacian

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Earnshaw's theorem

A scalar field $\phi(x, y)$ that has $\nabla^2 \phi = 0$ over a region in space, cannot have a local maxima or minima in that region.

At the best we can have a saddle point (say, $\frac{\partial^2 \phi}{\partial x^2} > 0$ and

$\frac{\partial^2 \phi}{\partial y^2} < 0$ making it feasible that $\nabla^2 \phi = 0$).

For any function of $\nabla^2 \phi = 0$, $D < 0$ $D = b_{xx}b_{yy} - b_{xy}^2$
i.e. only saddle point.

$$b_{xx} + b_{yy} = 0$$

i.e. if one is +ve then the other is -ve

$$D = b_{xx}b_{yy} - b_{xy}^2 < 0 \text{ i.e. saddle point}$$

Hyperbolic Paraboloid $\rightarrow \phi = x^2 - y^2$

$\rightarrow$ In an electrostatic field we can not hold a charge in a static equilibrium (due to Earnshaw theorem) whereas a time varying field can hold the same.

P.T.O

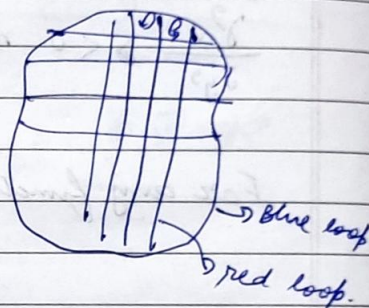
Curl Operator (~~Stoke's Theorem~~)

~~Curl of A is the~~

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad \text{lt} \quad \oint \vec{A} \cdot d\vec{s}$$

Stoke's Theorem

$$\int_{\text{Blue}} (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} = \oint_{\text{Blue}} \vec{A} \cdot d\vec{l}$$



When is this theorem not applicable?

- ↳ When direction of $d\vec{s}$ is not unique.
i.e. we can't tell what is inward or outward direction) i.e. in case of Mobius strip.

Irrrotational field :- (is a conservative field)

When curl of field is zero then it is irrotational field

i.e. $\vec{\nabla} \times \vec{F} = 0$: irrotational field.

Solenoidal Field:

i.e. divergence of a field is 0

$$\vec{\nabla} \cdot \vec{F} = 0$$

If $\vec{\nabla} \cdot \vec{F} > 0$ i.e. it is source of field

$\vec{\nabla} \cdot \vec{F} < 0$ i.e. " " sink of field

$\vec{\nabla} \cdot \vec{F} = 0$ i.e. no source and no sink. i.e. solenoidal field.

a Solenoidal field can always be written as $\vec{\nabla} \times \vec{F}$

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Q1)

$$\vec{F} = 3x^2 \hat{i} + 5xy^2 \hat{j} + xyz^3 \hat{k}$$

$$P = (1, 2, 3)$$

$$\nabla \cdot \vec{F} = ?$$

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\nabla \cdot \vec{F} = 6x \hat{i} + 10xy \hat{j} + 3yz^3 \hat{k}$$

$$\nabla \cdot \vec{F} = 18x^3 + 80x^2y^3 + 3x^2y^3z^5$$

$$= 18 + 80 \times 8 + 12 \times 3^5 = 3339$$

$$\nabla \cdot \vec{F} = 18 + 80 + 12 \times 3^5$$

$$\nabla \cdot \vec{F} = (6x + 10xy + 3yz^3)$$

$$= 6 + 20 + 3 \cdot 2 \cdot 9 = \boxed{80}$$

Q-2)

(a)

$$\vec{D} = r \sin \phi \hat{a}_r - \frac{1}{r} \sin \theta \cos \phi \hat{a}_\theta + r^2 \hat{a}_\phi$$

$$\vec{D} \text{ at } P(10, 150^\circ, 330^\circ)$$

$$r \quad \theta \quad \phi$$

$$= 10 \sin 330 \hat{a}_r - \frac{1}{10} \sin 150 \cos 330 \hat{a}_\theta + 100 \hat{a}_\phi$$

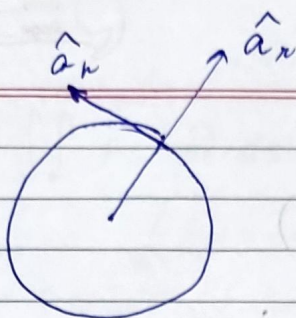
$$= \boxed{-5 \hat{a}_r - 0.0433 \hat{a}_\theta + 100 \hat{a}_\phi}$$

Tangent of cone
is along $\hat{a}_\theta$

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(b)



$$a = \hat{a}_n + \hat{a}_\theta$$

$$\hat{a}_n = -0.043 \hat{a}_\theta + 1.00 \hat{a}_\phi$$

(c) $\perp$ to $\vec{D}$ and tangential to cone ($\theta = 150^\circ$)

$\vec{u}$ is $\perp$ to both $\vec{D}$ and $\hat{a}_\theta$ as $\hat{a}_\theta$ is the tangent vector for cone.

$$\text{i.e. } \vec{u} = \vec{D} \times \hat{a}_\theta \Rightarrow = \begin{vmatrix} \hat{a}_n & \hat{a}_\theta & \hat{a}_\phi \\ -5 & 0.043 & 1.00 \\ 0 & 1 & 0 \end{vmatrix}$$

$$= -1.00 \hat{a}_n - 5 \hat{a}_\phi$$

$$\vec{u} = \frac{-1.00 \hat{a}_n - 5 \hat{a}_\phi}{\sqrt{1.00^2 + 5^2}}$$

Q-3) (a) $\vec{A} = 3 \hat{a}_n + 2 \hat{a}_\theta - 6 \hat{a}_\phi$
 $\vec{B} = 4 \hat{a}_n + 3 \hat{a}_\theta$

$$\vec{A} \cdot \vec{B} = 12 - 18 = -6$$

(b) $|\vec{A} \times \vec{B}| = \begin{vmatrix} \hat{a}_n & \hat{a}_\theta & \hat{a}_\phi \\ 3 & 2 & -6 \\ 4 & 0 & 3 \end{vmatrix} = 34.48$

Transformation of $\hat{a}_z \rightarrow \cos\theta \hat{a}_x + \sin\theta \hat{a}_y$
 $(\hat{k})$ ~~$\hat{a}_z \rightarrow \cos\theta \hat{a}_x + \sin\theta \hat{a}_y$~~

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$$C) = \vec{A} \cdot \hat{a}_z = ?$$

$$= \vec{A} \cdot (\cos\theta \hat{a}_x + \sin\theta \hat{a}_y)$$

$$= \vec{A} \cdot \left(\cos\frac{\pi}{3} \hat{a}_x + \sin\frac{\pi}{3} \hat{a}_y \right)$$

$$= (3\hat{a}_x + 2\hat{a}_y - 6\hat{a}_z) \cdot \left(\cos\frac{\pi}{3} \hat{a}_x + \sin\frac{\pi}{3} \hat{a}_y \right)$$

$$=$$

Q-4) $T = \ln(1/e)$

Div., grad
in spherical coordinates

Formula

$$F = - \vec{\nabla} T$$

$$= - \vec{\nabla} \ln(1/e)$$

$$= - \vec{\nabla} (-\ln e) = \vec{\nabla} \ln e$$

$$= \frac{\partial \ln e}{\partial \rho} \hat{a}_\rho$$

$$= \frac{1}{\rho} \hat{a}_\rho$$

$$\frac{\partial \ln e}{\partial x} \hat{i} + \frac{\partial \ln e}{\partial y} \hat{j}$$

$$+ \frac{\partial \ln e}{\partial z} \hat{k}$$

$$= \frac{1}{\rho} \frac{\partial e}{\partial x} \hat{i} + \frac{1}{\rho} \frac{\partial e}{\partial y} \hat{j}$$

$$+ \frac{1}{\rho} \frac{\partial e}{\partial z} \hat{k}$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

here $(\rho \text{ is } e)$

$$\iint \underbrace{\vec{F} \cdot \hat{n}}_d ds$$

$$= \iint \frac{1}{\epsilon_0} d\epsilon = \frac{1}{\epsilon_0} \iint ds$$

$$= \frac{1}{\epsilon_0} \cdot 4\pi \epsilon_0^2 = \underline{4\pi \epsilon_0}$$

$$\boxed{\left. \frac{\nabla \cdot \vec{F}}{\epsilon} \right| = \frac{1}{\epsilon^2} \frac{\partial}{\partial \epsilon} (\epsilon^2 \cdot F_\epsilon)} \quad (\text{Formula})$$

$$= \frac{1}{\epsilon^2} \frac{\partial}{\partial \epsilon} \left(\epsilon^2 \cdot \frac{1}{\epsilon} \right)$$

Q.5)

$$\vec{E} = -5 \hat{a}_\rho + 10 \hat{a}_\phi + 3 \hat{a}_z$$

cylindrical co-ordinates

$$P \equiv (5, \pi/9, 3)$$

$$\vec{F} = \hat{a}_\rho + 2 \hat{a}_\phi - 6 \hat{a}_z \quad (\text{Normal})$$

surface $z=3$

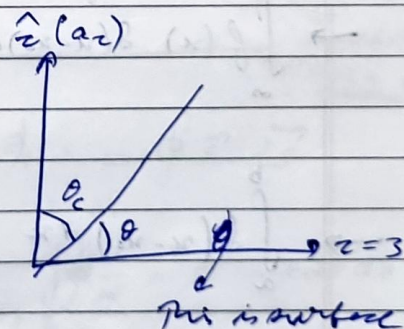
$$E \cdot a_z = |\vec{E}| \times \cos \theta_c = 3$$

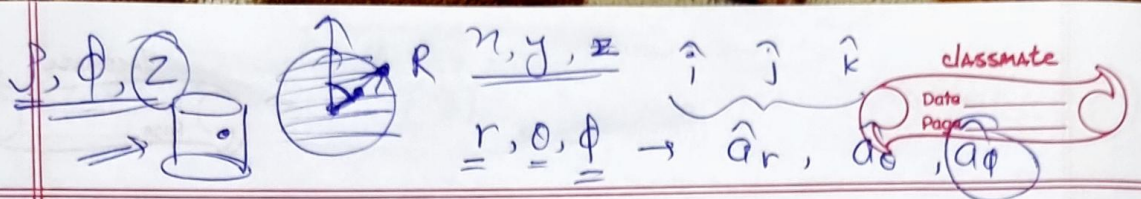
$$\sqrt{134} \cos \theta_c = 3$$

$$\cos \theta_c = \frac{3}{\sqrt{134}}$$

Hence θ is $90^\circ - \theta_c$

$$\text{Finall ans.} \rightarrow \underline{\theta = 15.02^\circ}$$





$$T = \ln(1/\rho)$$

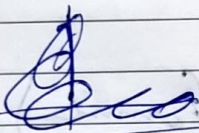
cy spke

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$Z = Z =$$

$$\begin{aligned} \nabla F &= \text{grad } F \Rightarrow \text{Car, spke, cyli} \\ \nabla \cdot F &= \text{Div } F \Rightarrow \end{aligned}$$

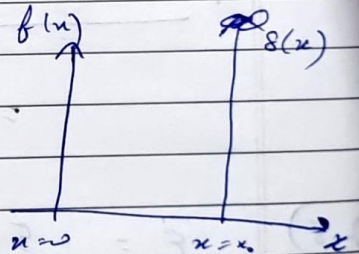
Curl



Ques-7

Dirac-Delta function:-

$$\left\{ \begin{aligned} \delta(x-x_0) &= \begin{cases} 0, & x \neq x_0 \\ \infty, & x = x_0 \end{cases} \\ \int_{-\infty}^{\infty} \delta(x-x_0) dx &= 1 \end{aligned} \right.$$



$$\begin{aligned} \rightarrow \int_{-\infty}^{\infty} f(x) \delta(x-x_0) dx &= \int_{-\infty}^{\infty} f(x_0) \delta(x-x_0) dx = f(x_0) \int_{-\infty}^{\infty} \delta(x-x_0) dx \\ &= f(x_0) \end{aligned}$$

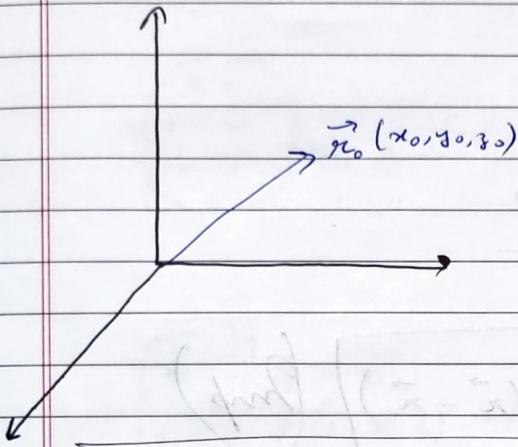
$$\rightarrow \int_a^b \delta(x-x_0) dx = \begin{cases} 1, & x \in [a, b] \\ 0, & x \notin [a, b] \end{cases}$$

$$\rightarrow \delta(kx) = \frac{1}{|k|} \delta(x)$$

$$\rightarrow \underline{\delta(x-x_0) = \delta(x-x)}$$

Delta function in 3-D

$$\delta^3(\vec{r} - \vec{r}_0) = \delta(x-x_0) \delta(y-y_0) \delta(z-z_0) = \begin{cases} 0, & r \neq r_0 \\ \infty, & r = r_0 \end{cases}$$



$$= \iiint_{-\infty}^{\infty} \delta^3(\vec{r} - \vec{r}_0) dV$$

$$= \int_{-\infty}^{\infty} \delta(x-x_0) dx \int_{-\infty}^{\infty} \delta(y-y_0) dy \int_{-\infty}^{\infty} \delta(z-z_0) dz = 1$$

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (F_\phi)$$

Ex: If $\vec{F} = \frac{\hat{r}}{r^2}$ find $\vec{\nabla} \cdot \vec{F}$

$$\vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = 0$$

Sol^y (M1) $\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \frac{1}{r^2} \right) = 0 \Rightarrow \int \vec{\nabla} \cdot \vec{F} dV = 0$

True for all points $\neq 0$.

(M2)

$$\int \vec{\nabla} \cdot \vec{F} dV \neq 0$$

$$\begin{aligned} \int_{-\infty}^{\infty} (\vec{\nabla} \cdot \vec{F}) dV &= \oint \vec{F} \cdot d\vec{S} \\ &= \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} \frac{\hat{r}}{r^2} \cdot \hat{r} r^2 \sin \theta d\theta d\phi \\ &= \int_0^{2\pi} d\phi \int_0^{\pi} \sin \theta d\theta = 4\pi \end{aligned}$$

This is non-zero divergence for $r=0$.
at every other point (except 0) divergence is 0 (M1)

$$\text{ie. } \int (\vec{\nabla} \cdot \vec{F}) dV = \begin{cases} 0, & r \neq 0 \\ 4\pi, & r = 0 \end{cases}$$

$$\vec{\nabla} \cdot \vec{F} = 4\pi \delta^3(r)$$

$$\vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(r)$$

$$\rightarrow \boxed{\vec{\nabla} \cdot \left(\frac{\hat{r}}{(r-r_0)^2} \right) = 4\pi \delta^3(\vec{r} - \vec{r}_0)} \quad (\text{Imp})$$

Ques) Find $\nabla^2 \left(\frac{1}{r} \right) = ?$ {Laplacian}

Soln

$$= \vec{\nabla} \cdot \vec{\nabla} \left(\frac{1}{r} \right)$$

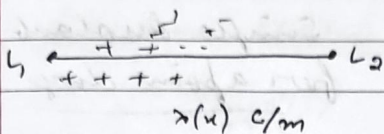
$$= - \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right)$$

$$= - 4\pi \delta^3(r)$$

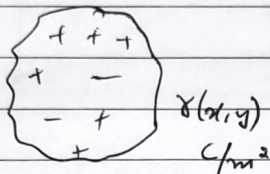
$$\vec{\nabla} \left(\frac{1}{r} \right) = \hat{r} \frac{d}{dr} \left(\frac{1}{r} \right)$$

$$= - \frac{\hat{r}}{r^2}$$

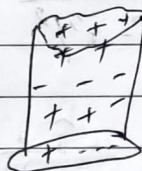
↓
gradient of $1/r$



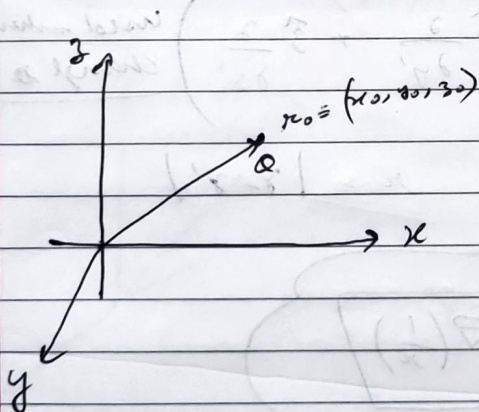
$$Q_{\text{total}} = \int_{L_1}^{L_2} \lambda(x) dx$$



$$Q_{\text{total}} = \int \gamma(x, y) dx dy$$


 $\rho(x, y, z)$
 $\rho(r, \phi, z)$
 $\rho(r, \theta, \phi)$
 C/m^3

$$Q_{\text{total}} = \int \rho(x, y, z) d\tau$$

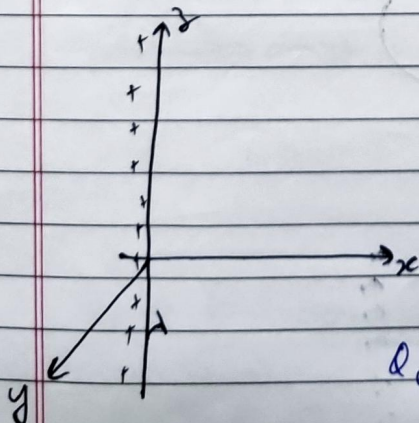


Volume charge density of a

point charge.

$$\rho = Q \delta^3(x - x_0)$$

$$Q_{\text{total}} = Q \int \delta^3(x - x_0) d\tau = Q$$



$$\int_{-\infty}^{\infty} \lambda dz = \iiint_{-\infty}^{\infty} \rho dx dy dz$$

$$= \int_{-\infty}^{\infty} \lambda dz \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(x) \delta(y) dx dy$$

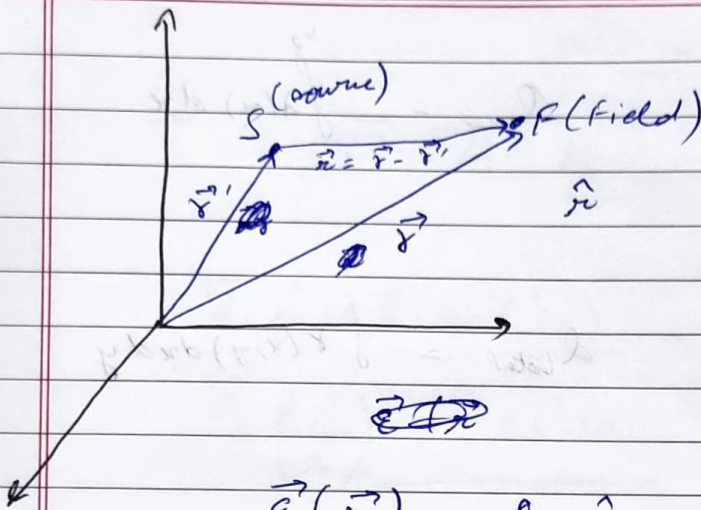
$$Q_{\text{total}} = \int_{-\infty}^{\infty} \lambda dz$$

$$\left(\begin{array}{l} \vec{r}' \rightarrow \text{Source} \\ \vec{r} \rightarrow \text{Field} \\ \vec{r} = (\vec{r} - \vec{r}') \text{ diff.} \end{array} \right)$$

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Simple Coulomb law for a point charge

$$\vec{E}(\vec{r}) = \frac{q}{4\pi\epsilon_0 r^2} \hat{r} = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{|\vec{r} - \vec{r}'|^2}$$

$$\vec{E} = -\vec{\nabla}\phi \rightarrow \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

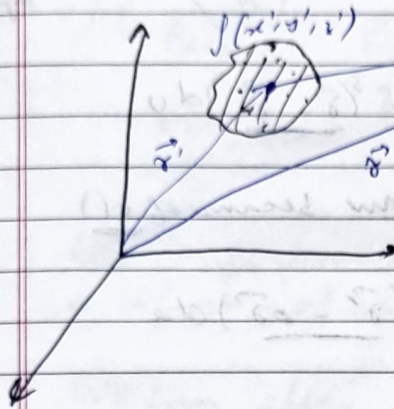
$$\left(\vec{\nabla}' = \hat{x} \frac{\partial}{\partial x'} + \hat{y} \frac{\partial}{\partial y'} + \hat{z} \frac{\partial}{\partial z'} \right) \text{ used when we change to } \vec{r}'$$

$$\vec{\nabla} F = \hat{r} \frac{\partial F}{\partial r} \frac{\partial r}{\partial x} \quad r = |\vec{r} - \vec{r}'|$$

If we put $\phi = -\frac{q}{4\pi\epsilon_0} \nabla\left(\frac{1}{r}\right)$

then $\vec{E} = -\vec{\nabla}\phi$

Coulomb charge for a volume charge



$$\vec{E} = \int \frac{\rho(x', y', z') dv' \hat{r}}{4\pi\epsilon_0 r^2}$$

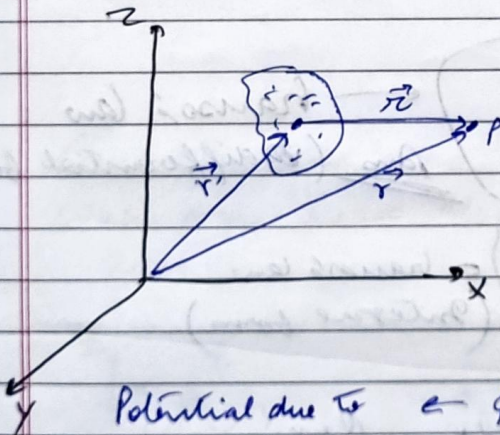
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \rho(x', y', z') dv' \frac{\hat{r}}{r^2}$$

Lect # 5

Recap:

$$\nabla \cdot \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(\vec{r})$$

$$= 4\pi \delta^3(\vec{r} - \vec{r}')$$



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{r}') dv' \left(\frac{\hat{r}}{r^2} \right)$$

↓
Electric field

$$\vec{E} = -\nabla\phi$$

Potential due to continuous charge distribution $\phi(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho dv'}{r}$

$$\nabla = \hat{r} \frac{\partial}{\partial r}$$

Divergence of $\vec{E}$ ($\nabla \cdot \vec{E}$)

$$\nabla \cdot \vec{E} = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\rho(\vec{r}') \frac{\hat{r}}{r^2} \right) dv'$$

$$= \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{r}') d\vec{r}' \vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right) \quad \left(\text{as } \vec{\nabla} \text{ only depends on } \vec{r} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{r}') d\vec{r}' \int \delta^3(\vec{r}' - \vec{r}) d\vec{r}$$

↓ saw because of $\delta()$

$$= \frac{1}{\epsilon_0} \int_V \rho(\vec{r}') \delta^3(\vec{r}' - \vec{r}) d\vec{r}$$

Using property

$$\left\{ \begin{aligned} f(x_0) &= \iiint_{-\infty}^{\infty} f(x) \delta^3(x - x_0) dx \\ \delta(\vec{r} - \vec{r}_0) &= \delta(\vec{r}_0 - \vec{r}) \quad \left(\text{even symmetry} \right) \end{aligned} \right\}$$

$$\boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho(\vec{r})}{\epsilon_0}} \quad \leftarrow \text{Gauss's law (differential form)}$$

$$\boxed{\oint \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enclosed}}}{\epsilon_0}} \quad \leftarrow \text{Gauss's law (Integral form)}$$

Proof Gauss differential to integral

$$\iiint (\vec{\nabla} \cdot \vec{E}) d\vec{r} = \frac{1}{\epsilon_0} \int \rho d\vec{r} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\text{Using formula } \Rightarrow \boxed{\int (\vec{\nabla} \cdot \vec{v}) d\vec{r} = \oint \vec{v} \cdot d\vec{s}}$$

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

do remove also !

$$\boxed{\nabla \times \vec{E} = 0} \quad \text{Curl of } \vec{E} \quad (\text{For a point charge})$$

Reason: $\vec{E}$ can be written as $\vec{E} = \vec{\nabla} \phi$ then $\nabla \times \vec{E} = 0$

$$\nabla \times \vec{E} = \nabla \times (\vec{\nabla} \phi) = 0 \quad \phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Use superposition principle to show that if we have a collection of charges then also $\nabla \times \vec{E} = 0$.

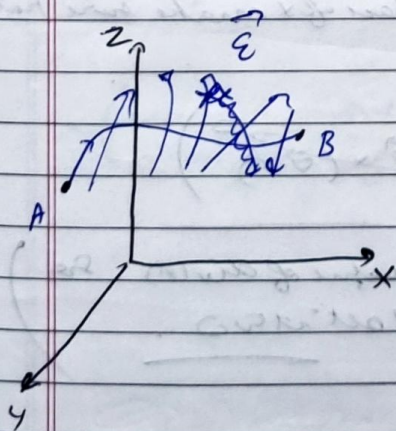
We can apply superposition because $E \propto q$ not q^2 or q^3 -

i.e.

$$\text{If, } \vec{E} = \sum_i \vec{E}_i \quad \nabla \times \vec{E} = \sum_i (\nabla \times \vec{E}_i) = 0$$

$$\boxed{\vec{E} = -\vec{\nabla} \phi} \rightarrow \text{Valid for all charges and any distribution.}$$

- Curl of E is zero only in case of electric field. When mag. field also comes in picture then it won't be zero.



$$\phi_{BA} = \phi_B - \phi_A$$

$$W = - \int_A^B \vec{E} \cdot d\vec{r}$$

$$= \int_A^B (\vec{\nabla} \phi) \cdot d\vec{r} = \int_A^B d\phi = \phi(B) - \phi(A)$$

(Formula)

Magnetic field ^(B) and potential (A)

$$\vec{B} = \nabla \times \vec{A}$$

$$\oint \vec{B} \cdot d\vec{s} = 0$$

$$\nabla \cdot \vec{B} = 0$$

similar to $\nabla \times \vec{E} = 0 \Rightarrow \vec{E} = -\nabla \phi$

If divergence of a any field is zero, then that field can be written as a curl of something.

mag. field divergence is 0, so it is a solenoidal field.

$$\vec{B} = \nabla \times \vec{A} \quad \text{similar to } \nabla \times \vec{E} = 0 \Rightarrow \vec{E} = -$$

$$\nabla \cdot \vec{B} = 0, \quad \vec{B} = (\nabla \times \vec{A})$$

$$= (\nabla \times \vec{A}_{new})$$

$$\vec{A}_{new} = \vec{A} + \nabla \times$$

(claim)

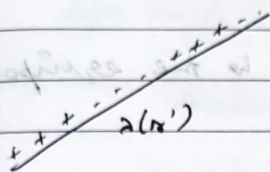
ref. point

Here we are writing $\vec{A}$ in some other vector but keeping in mind that $\vec{B}$ is same.

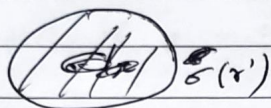
i.e. we can choose any thing in place of $\times$ make sure that $\vec{B}$ should be same.

$$\vec{B} = \nabla \times \vec{A}_{new} = \nabla \times \vec{A} + \nabla \times (\nabla \times \vec{A})$$

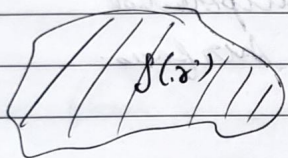
(Now we are free to choose any value of $\nabla \cdot \vec{A}$ for mathematical to be easier we will set it to 0.)



$$\phi(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(r')}{r} dr'$$

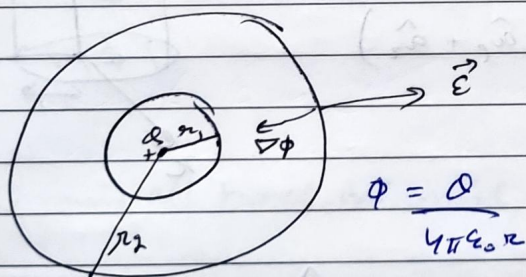


$$\phi(r) = \frac{1}{4\pi\epsilon_0} \iint \frac{\sigma(r')}{r} ds'$$



$$\phi(r) = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(r')}{r} dv'$$

level surfaces $\rightarrow$ equipotential surface.

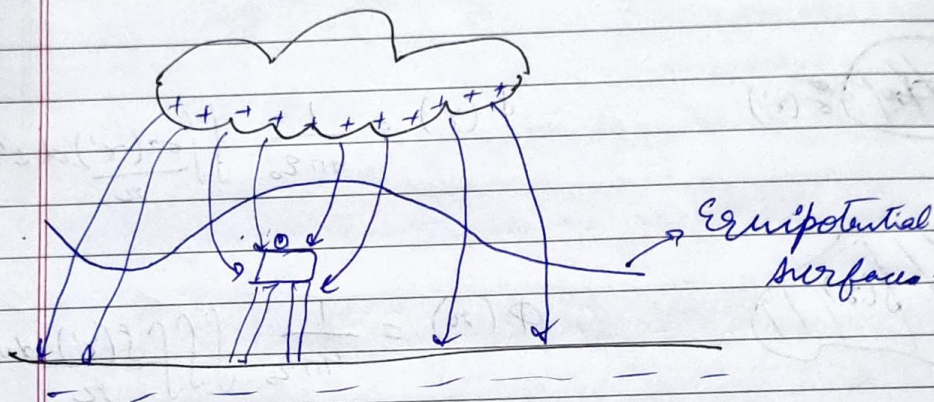


$$\phi(x, y, z) = \text{const.}$$

(as we move towards centre $r \downarrow$ so $\phi \uparrow$ i.e. direction of $\nabla \phi$ is inwards.)

P.T.O

- Electric field lines are always $\perp$ to the equipotential surfaces



Tut-2

Q1) $\vec{G}(r) = 10 e^{-2z} (\rho \hat{a}_\rho + \hat{a}_z)$

Flux = $\oint \vec{G} \cdot d\vec{s}$

for surface I

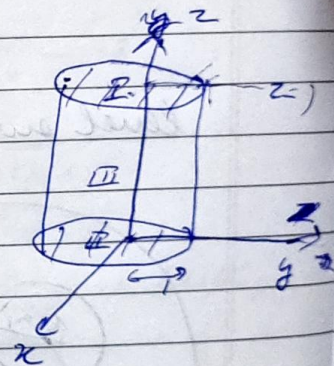
$$ds = r dr d\phi \hat{a}_z = \rho d\rho d\phi \hat{a}_z$$

$$\vec{G} \cdot d\vec{s} = (10 e^{-2z} (\rho \hat{a}_\rho + \hat{a}_z)) \cdot (\rho d\rho d\phi \hat{a}_z)$$

$$= 10 e^{-2z} \rho d\rho d\phi \quad (\text{here } z=1)$$

$$\text{Flux} = \int_{\phi=0}^{2\pi} \int_{\rho=0}^{5} 10 e^{-2} \rho d\rho d\phi = \int_{\phi=0}^{2\pi} 10 e^{-2} \left(\frac{\rho^2}{2} \right)_0^5 d\phi$$

$$= 50 e^{-2} (2\pi) = 10\pi e^{-2}$$



for surface II

Now here $z = 0$.

this is same as previous case. But here direction will be $-\hat{a}_z$.

$$G \cdot dS = 10 e^{-z} \int dS d\phi$$

Now here $z = 0$

$$\text{flux} = \iint 10 e^{0} \int dS d\phi$$

for surface III

Now for this surface z is varying instead of r .

summing all these, we must get 0.

By divergence theorem.

$$\oint \underbrace{\vec{G} \cdot d\vec{S}}_0 = \int \underbrace{(\nabla \cdot \vec{G})}_{\downarrow} dv$$

$$\nabla \cdot A = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\theta}{\partial \theta} + \frac{\partial A_z}{\partial z}$$

↳ formula of divergence of A in cylindrical coordinates

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Q.2) (a) $\vec{A} = x^2 y \hat{a}_x + y^2 z \hat{a}_y - 2xz \hat{a}_z$

Q-3)

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & y^2 z & -2xz \end{vmatrix}$$

$$= \hat{a}_x (0 - y^2) - \hat{a}_y (-2z - 0) + \hat{a}_z (0 - x^2)$$

$$= -y^2 \hat{a}_x + 2z \hat{a}_y - x^2 \hat{a}_z$$

$$\nabla \cdot (\nabla \times \vec{A}) = \frac{\partial}{\partial x} (-y^2) + \frac{\partial}{\partial y} (2z) + \frac{\partial}{\partial z} (-x^2)$$

$$= 0$$

(b) $\vec{A} = \rho^2 z \hat{a}_\rho + \rho^3 \hat{a}_\theta + 3\rho z^2 \hat{a}_z$

$$\nabla \times \vec{A} = \frac{1}{\rho} \begin{vmatrix} \hat{a}_\rho & \rho \hat{a}_\theta & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ \rho^2 z & \rho^3 & 3\rho z^2 \end{vmatrix} \quad (\text{Formula})$$

$$= \left(\frac{\rho^2 - 3z^2}{\rho} \right) \hat{a}_\rho + \left(\frac{4\rho^2}{\rho} \right) \hat{a}_\theta$$

$$\nabla \cdot (\nabla \times \vec{A}) = 0 \rightarrow (\text{Prove in exam})$$

$$\nabla \cdot \vec{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\theta}{\partial \theta} + \frac{\partial A_z}{\partial z}$$

Divergence of curl is 0

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Q-3) $\nabla \times \nabla V = 0$

$$\nabla \times \nabla V = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial y \partial z} \right) - \hat{j} \left(\frac{\partial^2 V}{\partial x \partial z} - \frac{\partial^2 V}{\partial x \partial z} \right) + \hat{k} \left(\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial x \partial y} \right)$$

$$= 0$$

Q4(a) $U = x^3 y^2 e^{xz}$ $P \equiv (1, -1, 1)$

$$\nabla^2 U = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) U$$

$$\frac{\partial U}{\partial x} = 3x^2 y^2 e^{xz} + x^3 y^2 e^{xz} z$$

$$\frac{\partial^2 U}{\partial x^2} = 6x y^2 e^{xz} + 3x^2 y^2 e^{xz} z + 3x^2 y^2 e^{xz} z + x^3 y^2 e^{xz} z^2$$

$$\frac{\partial^2 U}{\partial y^2} = 2x^3 e^{xz}$$

$$\frac{\partial^2 U}{\partial z^2} = x^3 y^2 e^{xz}$$

at $(1, -1, 1) \rightarrow$

$$\nabla^2 U = 6e + 3e + 3e + e + 2e + e = 16e$$

(b) $V = e^2 z (\cos \phi + \sin \phi)$ $P \equiv (5, \pi/6, -2)$

$$\nabla V = \frac{\partial V}{\partial e} \hat{e} + \frac{1}{e} \frac{\partial V}{\partial \phi} \hat{\phi} + \frac{\partial V}{\partial z} \hat{z} \quad \left. \vphantom{\frac{\partial V}{\partial e}} \right\} \text{Gradient of } V \text{ in cylindrical form.}$$

$$\nabla^2 V = \nabla \cdot (\nabla V)$$

$$\nabla V = 2e z (\cos \phi + \sin \phi) \hat{e} + \frac{1}{e} e^2 z (\cos \phi - \sin \phi) \hat{\phi} + e^2 (\cos \phi + \sin \phi) \hat{z}$$

formula of Laplacian in cylindrical co-ordinates

$$\nabla^2 V = \frac{1}{e} \frac{\partial}{\partial e} \left(e \frac{\partial V}{\partial e} \right) + \frac{1}{e^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

(use this here)

formula of Laplacian in spherical co-ordinates

$$\nabla^2 W = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial W}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial W}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 W}{\partial \phi^2}$$

Q5) $T = (\alpha xy + \beta z^3) \hat{i} + (3x^2 - \gamma z) \hat{j} + (3xz^2 - y) \hat{k}$

For irrotational field, $\vec{\nabla} \times \vec{T} = 0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \alpha xy + \beta z^3 & 3x^2 - \gamma z & 3xz^2 - y \end{vmatrix}$$

$$\hat{i}(-1 + \gamma) - \hat{j}(3z^2 - 3\beta z^2) + \hat{k}(6x - \alpha x) = 0$$

$$\alpha = 6 \quad \beta = 1, \quad \gamma = 1$$

$$\nabla \cdot \vec{T} = (\alpha y + 0 + 6xz) \quad \text{and } P \equiv (2, -1, 0)$$

$$= 6y + 6xz = -6 + 0 = \underline{-6}$$

P.T.O

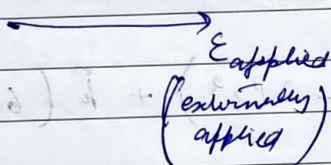
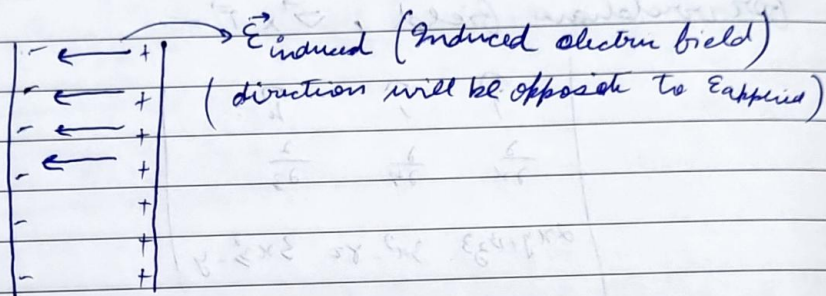
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lect #6

Perfect Conductor:- Infinite supply of e^- .



$$\vec{E}_{\text{total}} = \vec{E}_{\text{applied}} + \vec{E}_{\text{induced}}$$

constant

increases till it becomes equal to $\vec{E}_{\text{applied}}$ and $\vec{E}_{\text{total}} = 0$.

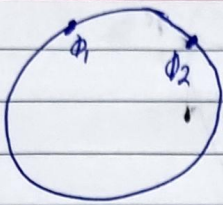
① → ~~The process of~~ In perfect conductor, this process is instantaneous. So the perfect conductor has 'zero' electric field inside.

② → The tangential $\vec{E}$ ~~induced~~ at surface of perfect electric conductor (PEC) is always zero. (because if it is there ~~this charge~~ charge redistribution again takes place and there is $\vec{E}$ inside PEC which violates first point)

→ $\vec{E}$ normal across the surface of PEC is not necessarily zero.

→ its surface of a PEC is equipotential.

(3)*



ϕ_1 and ϕ_2 both are at same potential as tangential comp of both are zero i.e. Equipotential surface.

(4)

Volume charge density of a PEC is always 0.

$$\vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Rightarrow \boxed{\rho = 0}$$

$\Rightarrow \vec{E} = 0$ inside a conductor

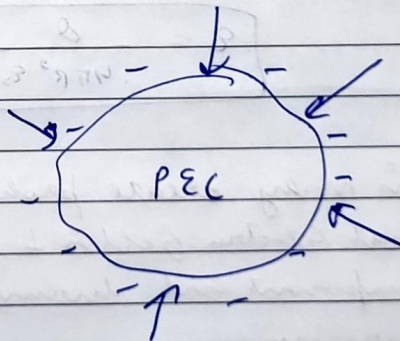
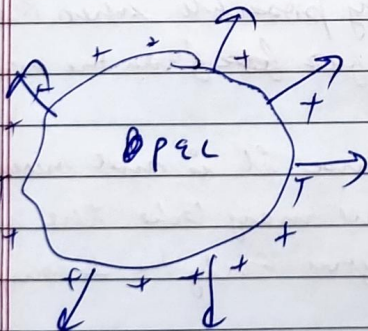
$$\Rightarrow \boxed{\rho = 0}$$

(5)*

So if we forcefully place a charge inside the PEC and when we apply ~~the~~ external electric field then due to induced ~~the~~ $\vec{E}$ the forceful charge gets distributed to the surface.

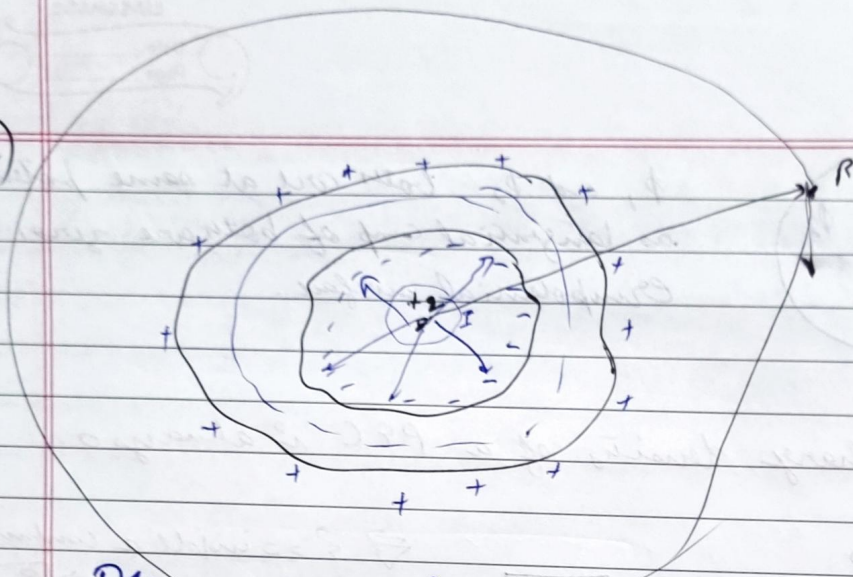
(6)*

~~Due~~ If the shape of conductor is random, then the redistribution will happen till the tangential comp of $\vec{E}$ becomes 0.



P.T.O

(1)



The $+q$ charge induces $-q$ charge inside the cavity. There will be field inside the cavity but not inside the conductor.

Electric field at any point R :-

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

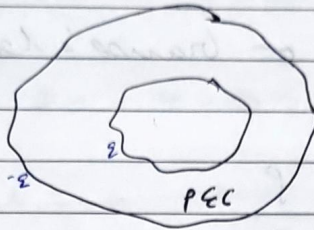
$$E(4\pi R^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi R^2 \epsilon_0}$$

(Only possible when R is very far from the surface)

Note:- This may create problem because it is not necessary that electric field is $\perp$. But we may take the component and during the integral only the $\perp$ comp. survives.

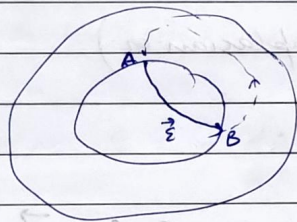
(II)

 $Q.$ 

- Q-1) What is the $\vec{E}$ on surface of PCC?
- Q-2) What is the $\vec{E}$ inside the shell?

Solⁿ

There will be no $\vec{E}$ inside PCC as well as cavity.

Proof:-

We can assume no $\vec{E}$ inside cavity.

We know $\Rightarrow \oint \vec{E} \cdot d\vec{r} = 0$

$$\int_A^B \vec{E} \cdot d\vec{r} + \int_B^A \vec{E} \cdot d\vec{r} = 0 \quad \left(\text{as } \vec{E} \text{ inside PCC is } 0 \right)$$

$$\text{So } \int_A^B \vec{E} \cdot d\vec{r} = 0 \quad \text{Now for this to happen } \boxed{\vec{E} = 0}$$

i.e. we do not have any $\vec{E}$ inside the cavity.

This is called Faraday Cage

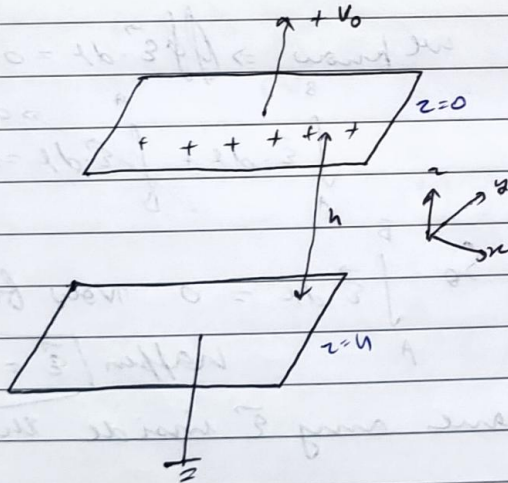
Poisson's equation

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \leftarrow \text{Gauss's law}$$

$$\Rightarrow \nabla \cdot (-\nabla \phi) = \frac{\rho}{\epsilon_0}$$

$$\boxed{\nabla^2 \phi = -\frac{\rho}{\epsilon_0}} \quad (\text{Poisson's eq.})$$

$$\text{If } \boxed{\rho = 0} \quad \boxed{\nabla^2 \phi = 0} \quad (\text{Laplacian eq.})$$

QuesFind $\vec{E}$.

$$\text{Sol}^n \quad \frac{d^2 \phi}{dz^2} = 0 \quad (\text{Laplacian eq.})$$

$$\Rightarrow \frac{d\phi}{dz} = C_1$$

$$\boxed{\phi = C_1 z + C_2}$$

Now, when $z=0$ $V_0 = C_0$

$$\text{i.e. } \phi = C_1 z + V_0$$

$$\phi \text{ at } z=h, \phi=0$$

$$0 = C_1 h + V_0 \Rightarrow \boxed{C_1 = -\frac{V_0}{h}}$$

$$\text{i.e. } \boxed{\phi = -\frac{V_0}{h} z + V_0}$$

$$\text{We know, } \epsilon = -\frac{d\phi}{dz} = \underline{\underline{\frac{V_0}{h}}}$$

Ans) Now, the question is why is the ϕ only a function of z not x, y ?

Solⁿ Because the structure is continuous translational symmetric.

Q1) (a) $\int_2^4 (6x^2 - 3x + 2) \delta(x-2) dx$

$$= (6(2)^2 - 3(2) + 2) \int_2^4 \delta(x-2) dx$$

$$= 24 - 6 + 2 = 20$$

(b) $\int_{-\infty}^{\infty} \ln(x+1) \delta(x-2) dx = \ln(3)$

Q2) $\boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$

(a) $\vec{E} = k r^3 \hat{r}$

$$\boxed{\vec{\nabla} \cdot \vec{r} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin \theta \cdot 0) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (r \sin \theta \cdot 0)}$$

here $\vec{\nabla} \cdot \vec{E} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot k r^3)$

$$= \frac{1}{r^2} \cdot k \cdot 5 r^2 = 5 k r^2$$

$$\rho = 5 k r^2 \epsilon_0$$

(b) ~~Q3)~~ $V = \frac{4}{3} \pi r^3$

$$dV = 4 \pi r^2 dr$$

$$\text{Total charge} = \int \rho dV = \int_0^R 5 k r^2 \epsilon_0 \cdot 4 \pi r^2 dr$$

$$= \left(4 k \pi \epsilon_0 r^5 \right)_0^R = \boxed{4 k \pi R^5 \epsilon_0}$$

OR

going $\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0} \Rightarrow \int \vec{\nabla} \cdot \vec{E} dV = \frac{q_{enc}}{\epsilon_0}$

div. theorem

If ρ was const. then total charge would be ($\rho \cdot V$)

Q-3)

$$\oint \epsilon_0 \vec{E} \cdot d\vec{s} = q_{enc} = \int \rho_v dV$$

$$\epsilon_0 \vec{E}_r \cdot \underbrace{4\pi r^2}_{\uparrow d\vec{s}} = \int_0^r \underbrace{4\pi r^2 \frac{\rho_0 r}{R}}_{\downarrow} dr$$

$$= \frac{\rho_0 \pi r^4}{R}$$

~~Ans~~
Ans $\rightarrow$

$$\vec{E}_r = \begin{cases} \frac{\rho_0 r^2}{4\epsilon_0 R} \hat{a}_r, & 0 \leq r \leq R \\ \frac{\rho_0 R^3}{4\epsilon_0 r^2} \hat{a}_r, & r > R \end{cases}$$

Q4)

(m1)

$$\vec{\nabla} \left(\frac{\hat{r}}{r^2} \right) = 4\pi \delta^3(r) \quad (\text{Formula})$$

Confusion?

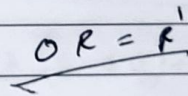
$$\vec{F} = \int (r^2 + 2) 4\pi \delta^3(r) dz = (0+2) 4\pi \int \delta^3(r) dz$$

$$= 8\pi$$

~~(Not a student for entry)~~

$$2\psi \rightarrow 8\pi$$

$$-\int_{\gamma} \frac{1}{z} \cdot (2\pi) dz = 2\pi \int_{\gamma} \frac{1}{z} dz$$



potential at R in terms of r_1 and r_2 .

$$V_R = \frac{k(-a)}{r_2} + \frac{kq}{r_1} = \frac{kq}{r_1} - \frac{kq}{r_2} = kq \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Now - $DR \approx AR = \pi_1$
 $CR \approx BR = \pi_2$

für $\pi', \pi_1, \pi_2 \gg d$

$$OD = OQ = \frac{d}{2} \cos \theta$$

$$r_1 = R' + \frac{d}{2} \cos \theta$$

$$r_2 = R' - \frac{d}{2} \cos \theta$$

put in (1)

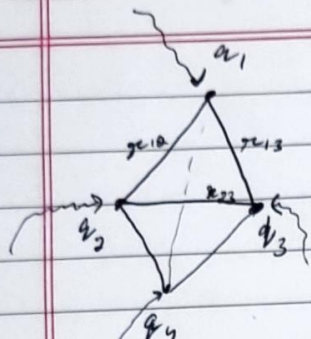
$$V_P = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{R' + \frac{d}{2} \cos \theta} - \frac{1}{R' - \frac{d}{2} \cos \theta} \right)$$

$$= \frac{q}{4\pi\epsilon_0} \frac{-d \cos \theta}{(R')^2 - \frac{d^2}{4} \cos^2 \theta}$$

$$= \frac{q}{4\pi\epsilon_0} \frac{d \cos \theta}{(R')^2} \quad \{ qd = p \} \text{ dipole moment}$$

$$V_{\text{at } P} = \frac{-q}{4\pi\epsilon_0} \left(\frac{p \cos \theta}{(R')^2} \right)$$

(7)



We firstly bring q_1 from ∞
then q_2 then q_3 then q_4 .

$$W_1 = 0$$

$$W_2 = \frac{1}{4\pi\epsilon_0} q_2 \frac{q_1}{r_{12}}$$

$$W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_2}{r_{23}} + \frac{q_1}{r_{13}} \right)$$

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_3}{r_{34}} + \frac{q_2}{r_{24}} + \frac{q_1}{r_{14}} \right)$$

$$W = W_1 + W_2 + W_3 + W_4 + \dots + W_i + \dots = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{j < i} \frac{q_i q_j}{r_{ij}}$$

• This would be practically not possible as it will be very calculative.

• In the above formula, order matters as which charge is brought first.

• To get rid of the condition $[j < i]$ we will deliberately calculate double the value and then put a half sign before.

i.e

$$W = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^N q_i \sum_{\substack{j=1 \\ (i+1)}}^N \frac{q_j}{r_{ij}}$$

~~Principle~~ • In this formula order does not matter.

• Also we won't consider the effect from the charge itself.
i.e. of

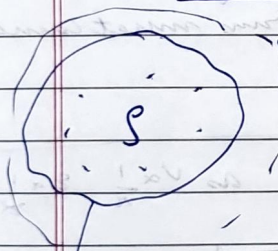
$$= \frac{1}{2} \sum_{i=1}^N q_i \left\{ \sum_{j=1}^N \frac{q_j}{4\pi\epsilon_0 r_{ij}} \right\}$$

→ Potential at a_i due to all other charges

$$W = \frac{1}{2} \sum_{i=1}^N q_i \underline{V(r_i)}$$

Potential at the ~~point~~ position of i^{th} charge due to all other charges.

For discrete charge distribution



$$dW = \frac{1}{2} \rho v dv$$

$$dW = \frac{1}{2} \int \rho v dv \text{ over all space}$$

→ works for all space.

$$W = \frac{1}{2} \int \rho v dv$$

→ If we choose a bigger volume like this then outside boundary $S=0$ i.e. same W. (does not matter)

→ Now, we will try to relate it with electric field. (for simplicity) (next page)

Note: • $\nabla \cdot (f \vec{A}) = f (\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla f)$

• $\int \nabla \cdot (f \vec{A}) dV = \int f (\nabla \cdot \vec{A}) dV + \int \vec{A} \cdot (\nabla f) dV$

$\Rightarrow \oint f \vec{A} \cdot d\vec{s} - \int \vec{A} \cdot (\nabla f) dV = \int f (\nabla \cdot \vec{A}) dV$

S_1 : surface area of this

classmate

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Page _____

Using Gauss's law

$$\oint w = \frac{\epsilon_0}{2} \int (\nabla \cdot \vec{E}) v d\tau$$

(Using previous formula)

$$= \frac{\epsilon_0}{2} \oint_{S_1} v \vec{E} \cdot d\vec{s} - \int_{V_1} \vec{E} \cdot (\nabla v) d\tau$$

$$= \frac{\epsilon_0}{2} \left(\oint_{S_1} v \vec{E} \cdot d\vec{s} + \int_{V_1} |\vec{E}|^2 d\tau \right)$$

Now we need to justify that this converges as $\pi \rightarrow \infty$. i.e. the terms must converge as $\pi \rightarrow \infty$.

Now integral I is converging as $\pi \rightarrow \infty$ as $v \propto \frac{1}{\pi}$ & $\epsilon \propto \frac{1}{\pi^2}$

$$ds \propto \pi^2$$

$$\text{i.e. } \oint v \vec{E} \cdot d\vec{s} \propto \frac{1}{\pi}$$

$$\text{i.e. } \frac{\epsilon_0}{2} \oint_{\text{all space}} v \vec{E} \cdot d\vec{s} \rightarrow 0 \text{ as } \pi \rightarrow \infty$$

Now, we are left with other term

$$\text{as } \vec{E} \propto \frac{1}{\pi^2}, |\vec{E}|^2 \propto \frac{1}{\pi^4}, v \propto \pi^3$$

$$\text{i.e. } |\vec{E}|^2 dV \propto \frac{1}{\pi} \text{ i.e. it also converges}$$

as $|\vec{E}|^2 > 0$, the above is converging

$$\text{i.e. } \frac{\epsilon_0}{2} \oint_{\text{all space}} |\vec{E}|^2 d\tau$$

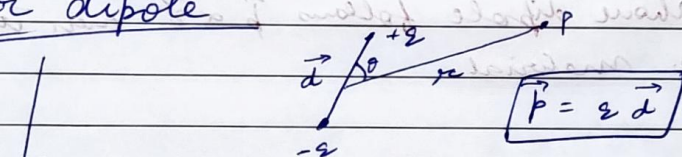
i.e. first term tends to 0
and second term is converging but not 0.

i.e.

$$W = \frac{\epsilon_0}{2} \int_{\text{all space}} |\vec{E}|^2 d\tau \quad (\text{Ran})$$

(Electrostatic energy density)

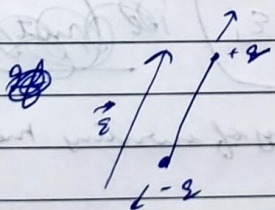
For dipole



$\vec{E} \propto \frac{1}{r^3}$, $V \propto \frac{1}{r^2}$ i.e. the above integral converges much faster.

Potential at point P

$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$



If $\vec{E}$ is applied as given, then $+q$ goes away i.e. $p \uparrow$ i.e. $V \uparrow$.

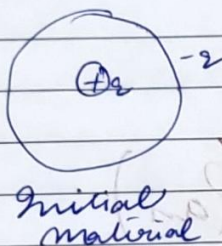
i.e. $\vec{p} = \alpha \vec{E}$ (For linear materials)

$\alpha > 0$ i.e. $\vec{p}$ and $\vec{E}$ are in same direction.

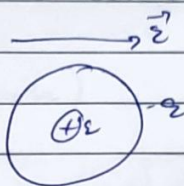
Majority we will be dealing with this (linear material only)

Ques) why the dipole moment (previous page) linp?

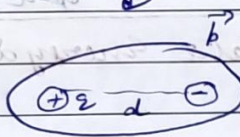
Solⁿ



apply $\vec{E}$



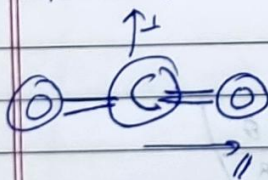
Effect of $\vec{E}$



(This is called induced dipole moment)

If the above dipole follows $\vec{p} \propto \vec{E}$ then it is linear material.

Suppose we have a CO_2 molecule and we apply $\vec{E}$ both $\perp$ and $\parallel$. $\parallel$ ~~comp~~ $\vec{E}$ will be more effective for the separation of charges.



$$p_{\parallel} = \alpha_{\parallel} E_{\parallel}, \quad p_{\perp} = \alpha_{\perp} E_{\perp}$$

$$\vec{p} = \alpha_{\parallel} E_{\parallel} + \alpha_{\perp} E_{\perp} \quad \text{or} \quad \vec{p} = \alpha \vec{E}$$

→ For 3-D molecules, most general way of writing the induced dipole moment is

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix}$$

induced dipole moment

polarizability tensor (matrix)

applied $\vec{E}$

For isotropic material (linear material)

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_z \end{bmatrix}$$

$$p_x = \alpha \epsilon_x$$

$$p_y = \alpha \epsilon_y$$

$$p_z = \alpha \epsilon_z$$

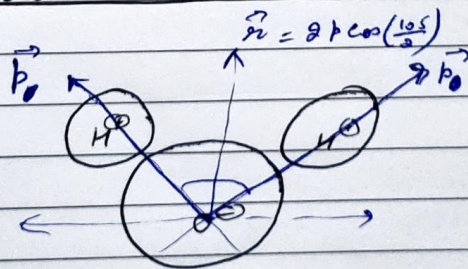
In vector form $\Rightarrow \vec{p} = \hat{x} p_x + \hat{y} p_y + \hat{z} p_z$

$$= \alpha \epsilon_x \hat{x} + \alpha \epsilon_y \hat{y} + \alpha \epsilon_z \hat{z}$$

$$= \alpha (\hat{x} \epsilon_x + \hat{y} \epsilon_y + \hat{z} \epsilon_z)$$

$$\boxed{\vec{p} = \alpha \vec{\epsilon}}$$

H₂O molecule: Polar molecule



$\vec{p} \rightarrow$ dipole

- For haphazardly arranged dipoles, when we apply $\vec{\epsilon}$ then they will align in a particular way. (Not perfectly aligned due to thermal fluctuations)
- For induced dipole this already exists.